THE WORDS AND WHAT THEY MEAN

Securement has its own vocabulary, and the regulations use it without introduction. Nothing here is complicated once someone says it plainly. Flip back to this page whenever a term in the guide is doing work you cannot see.

A THE RATINGS, AND THE ONE THAT COUNTS

- **WORKING LOAD LIMIT (WLL) — WHAT A DEVICE MAY BE WORKED AT**
  The load a strap, chain, or fitting is rated to carry in normal use. Every calculation in the securement rules runs on this number and no other.
- **BREAKING STRENGTH — WHERE IT FAILS**
  The point at which the device actually breaks. Always a much larger number than the working load limit, and a device advertised by this figure is quoting the one that does not count.
- **AGGREGATE WORKING LOAD LIMIT — THE TOTAL ACROSS ALL TIEDOWNS**
  Add up what each tiedown contributes, counted by the rules in Part One. "Aggregate" simply means the sum.
- **G — A MULTIPLE OF THE LOAD'S OWN WEIGHT**
  One g is the force of gravity. So 0.8 g forward means a force equal to **80 percent of the cargo's weight**, thrown forward in a hard stop. On a 20,000 lb load that is 16,000 lb of shove. This is why the rules concentrate on the front of a load.
- **DECELERATION — SLOWING DOWN**
  Acceleration in reverse. The forward figures in the performance criteria describe braking; the rearward and lateral figures describe pulling away and cornering.

B THE HARDWARE

- **TIEDOWN — THE WHOLE ASSEMBLY, NOT JUST THE STRAP**
  A chain, strap, rope, wire rope, or steel band, together with every hook, fitting, and tensioner in it. Its rating is that of its weakest part.

- **ANCHOR POINT — WHERE A TIEDOWN ATTACHES**
  A rated attachment on the vehicle, such as a rub rail pocket or D-ring, or a designed attachment on the cargo itself. Anchor points carry their own working load limit, and a weak one caps the whole tiedown.

- **BINDER — THE TIGHTENER USED WITH CHAIN**
  A lever or ratchet device that takes the slack out of a chain. It has its own rating, and it is very often the weakest component in a chain assembly.

- **TENSIONER — THE GENERAL TERM**
  Anything that tightens a tiedown — a binder on chain, a ratchet on webbing, a winch on a flatbed. The regulation names it specifically when defining the weakest-link rule.

- **EDGE PROTECTION — WHAT GOES BETWEEN THE STRAP AND THE CORNER**
  A sleeve, corner, or guard that stops a tiedown being cut or worn where it crosses cargo. It must resist abrasion, cutting **and crushing**, which is why folded cardboard does not qualify.

- **FRICTION MAT — A RUBBER PAD THAT RESISTS SLIDING**
  Placed under cargo to stop it sliding. Unrated mats count as providing resistance equal to 50 percent of the weight sitting on them.

C HOLDING CARGO IN PLACE WITHOUT STRAPS

- **BLOCKING — STOPPING MOVEMENT IN ONE DIRECTION**
  Something placed against the cargo so it cannot slide that way. Timber against the front of a crate is blocking.

- **BRACING — HOLDING IT FROM THE OTHER SIDE**
  Structure that pushes back against the cargo, usually working with blocking to pin it in place.

- **DUNNAGE — LOOSE MATERIAL USED TO FILL AND PROTECT**
  Timber, boards, or inflatable dunnage bags placed between or around cargo to take up space and stop movement. Damaged dunnage that would compromise the system is not permitted.

- **SHORING BAR — AN ADJUSTABLE BAR ACROSS A TRAILER**
  Braced between the walls of a van to hold a load or fill a void behind it.
- **CHOCK AND WEDGE — FOR ANYTHING THAT ROLLS**
  Tapered blocks placed against a round object so it cannot roll. They must be held in place by something other than friction.
- **CRADLE — A SHAPED SUPPORT**
  A fitting that holds a round item in a seat so it cannot roll, supporting it off the deck.
- **SPACER — WHAT SEPARATES STACKED TIERS**
  Material placed between layers of bundled cargo, with its own rules on width and support.
- **VOID — EMPTY SPACE IN THE LOAD**
  A gap the cargo could move into. Voids are the reason partial loads in vans fail when full loads do not.

D PARTS OF THE VEHICLE

Plain descriptions. The formal legal definitions sit in 49 CFR 390.5 and 393.5 if you need the exact wording.

- **HEADERBOARD AND BULKHEAD — THE WALL AT THE FRONT OF THE DECK**
  The structure between the cargo and the cab, sometimes called a headache rack. It has its own height, width, strength, and penetration requirements, because it is what stands between a shifting load and the driver.
- **STAKE AND STANDARD — UPRIGHTS ALONG THE SIDE**
  Vertical posts fitted into pockets on a flatbed to contain a load sideways.
- **BUNK AND BOLSTER — CROSS SUPPORTS THAT CRADLE A LOAD**
  Transverse structures, most often on log trailers, that hold a load and stop it rolling off.
- **SIDED VEHICLE — ANYTHING WITH WALLS**
  A van or box body. Walls of adequate strength are themselves a securement method when the load is tight against them.
- **FLATBED, LOWBOY, CURTAIN-SIDED — OPEN DECKS**
  A flatbed is an open deck. A lowboy sits low for tall or heavy equipment. A curtain-sided trailer has fabric sides, which are weather protection rather than securement.

- **FULL TRAILER AND POLE TRAILER — TWO TRAILER TYPES NAMED IN THE RULES**
  A full trailer carries its own weight on its own axles rather than resting on the tractor. A pole trailer is a bare frame connected by a reach or pole, used for long items like poles and logs.

- **CONVERTER DOLLY — THE AXLE UNIT THAT MAKES A SECOND TRAILER POSSIBLE**
  A coupling unit between trailers. It counts as its own unit for annual inspection purposes.

- **INTERMODAL CONTAINER — A SHIPPING CONTAINER THAT MOVES BETWEEN SHIP, RAIL, AND ROAD**
  The steel box itself, as distinct from the trailer carrying it. It has its own securement section, and cargo **inside** it is secured under the ordinary rules.

- **CONTAINER CHASSIS — A FRAME BUILT TO CARRY A SHIPPING CONTAINER**
  Fitted with locking devices at the corners rather than a deck.

E MATERIALS AND MARKINGS

- **GRADE — A CHAIN'S STRENGTH CLASS**
  Stamped on the links. Grade 30 proof coil is the weakest, then Grade 43 high test, Grade 70 transport, and the alloy grades 80 and 100. Unmarked chain is treated as Grade 30.

- **PROOF COIL — THE COMMON NAME FOR GRADE 30**
  General-purpose chain, not intended as transport tiedown chain. It is also the fallback rating for any chain you cannot identify.

- **ALLOY — HIGHER-STRENGTH CHAIN**
  Grades 80 and 100. Usually stronger, though at 5/16 inch Grade 80 is rated slightly below Grade 70 in the federal table.

- **WEBBING — FLAT SYNTHETIC STRAP**
  What most people mean by "straps". Rated by width.

- **CORDAGE — THE REGULATION'S WORD FOR ROPE**
  Manila, polypropylene, polyester, and nylon. Rope that cannot be identified by composition is rated as polypropylene, the weakest of them.

- **WIRE ROPE AND 6 × 37 FIBER CORE — STEEL CABLE, AND ITS CONSTRUCTION**
  "6 × 37" describes how the strands are laid up; "fiber core" means the centre is fibre rather than steel. Wire rope whose construction cannot be identified is rated as this type.

- **STEEL STRAPPING, SEAL, CRIMP, LAP JOINT — BANDING AND HOW IT IS JOINED**
  Steel band held by a metal sleeve called a seal, squeezed shut with crimps. A lap joint is where the band overlaps itself end over end. Both the crimp count and the seal count are specified.

F ROUND CARGO: COILS, ROLLS, AND PIPE

- **EYES — THE HOLE THROUGH THE MIDDLE**
  Eyes vertical: the hole points at the sky and the coil sits like a doughnut on a table. **Eyes crosswise:** the hole points at the sides of the trailer, so it would roll forwards and backwards. **Eyes lengthwise:** the hole points forward, so it would roll side to side. Which one you have decides which rules apply.

- **COIL BUNK — A FITTING THAT HOLDS CHOCKS AND TIMBERS IN PLACE**
  A device on the deck that keeps the blocking under a coil from working loose. Required where timbers, chocks or wedges are used, because nails alone are not permitted.

- **CLEAT — A SMALL NAILED STRIP**
  A wooden or metal piece fastened to the deck to stop something sliding. Nailed cleats may not be the sole means of securing chocks or timbers under a coil.

- **WELL — THE TROUGH BETWEEN TWO ROUND ITEMS**
  The natural seat formed where two rolls or pipes sit side by side. An upper layer is built by placing items into these wells, and several stacking rules turn on whether every well beneath is filled.

- **BLOCKING ROLL — A DELIBERATELY TALLER ROLL USED AS A STOP**
  A paper roll set higher than its neighbours — either taller by nature or raised on dunnage — so it blocks the rolls in the layer above from moving forward.

- **BANDING — STRAPPING ITEMS TOGETHER INTO ONE GROUP**
  Bands applied around several rolls so they behave as a single stable unit. Bands must be tight and secured so they cannot fall off the rolls or onto the deck.

- **UNITIZED — MADE INTO ONE ARTICLE**
  Bundled or banded so a group of items is treated as a single article of cargo. Unitized loads often fall back to the general rules rather than a commodity-specific section.

- **VOID FILLER — ANYTHING THAT TAKES UP EMPTY SPACE**
  Material placed in a gap so cargo cannot move into it. Airbags, timber, or packing all serve.

G LOGS, LUMBER, AND PIPE

- **SHORTWOOD AND LONGWOOD — LOG LENGTH CATEGORIES**
  Shortwood is cut log, handled in stacks and often loaded crosswise. Longwood is full-length timber, which must be cradled in two or more bunks. The securement rules differ between them.

- **PROCESSED LOG — A LOG THAT HAS BEEN WORKED**
  Squared, sawn or otherwise shaped rather than left in the round. A load of no more than four of them falls outside the log-specific rules.

- **CRIB-TYPE LOG TRAILER — A TRAILER THAT BOXES THE LOAD IN**
  A trailer whose structure contains the logs on its own. Correctly loaded, it is the one case where tiedowns are not additionally required. The formal definition is at 49 CFR 393.5.

- **WRAPPER — A TIEDOWN THAT ENCIRCLES THE WHOLE LOAD**
  Wire rope or similar passed right around a stack to bind it together. Where a wrapper is bundling logs rather than holding them to the truck, it need not be attached to the vehicle.

- **DRESSED LUMBER — MILLED, FINISHED TIMBER**
  Planed to size, as opposed to rough-sawn. The lumber section also covers packaged lumber and building products of similar shape, such as plywood and gypsum board — the plasterboard sold in large flat sheets.

- **HIGH-FRICTION DEVICE — A GRIP LAYER BETWEEN TIERS**
  Material placed between stacked bundles to stop them sliding on each other, used as an alternative to stakes for lateral restraint.

- **BELL PIPE — PIPE WITH A FLARED END**
  One end is widened so the next pipe socket into it. The flare has to be kept clear of the deck on spacers, and the bells alternate sides in a way the regulation sets out precisely.

- **PIPE UNLOADER — THE HANDLING ARM ON THE TRAILER**
  Equipment used to load and unload pipe. When locked, it counts as a means of immobilising the front or rear pipe.

H VEHICLES AND MACHINERY AS CARGO

- **MOUNTING POINT — THE TIEDOWN EYE BUILT INTO A VEHICLE**
  A fixing designed by the manufacturer to be strapped to. Where a tiedown attaches to the vehicle's structure it must use these, rather than a bumper, axle, or suspension part.

- **ARTICULATED — HINGED IN THE MIDDLE**
  A machine whose front and rear halves pivot against each other, such as an articulated loader. It must be restrained so it cannot fold while being carried.

- **CRAWLER TRACK — THE CONTINUOUS BELT ON A TRACKED MACHINE**
  What an excavator or bulldozer runs on instead of wheels. Tracked and wheeled machines are treated identically by the securement rules.

- **HYDRAULIC SHOVEL, FRONT END LOADER, POWER SHOVEL — MACHINES NAMED IN THE REGULATION**
  Examples the section gives of heavy equipment. What matters is not the name but that the machine operates on wheels or tracks and weighs 10,000 lb or more.

- **ACCESSORY EQUIPMENT — THE MOVING PARTS OF A MACHINE**
  Buckets, booms, blades and arms. All of it must be completely lowered and secured to the transporting vehicle before travel, because hydraulics bleed down on the road.

- **VEHICLE STACK — A PILE OF CRUSHED CARS**
  The unit the scrap rules count against. Tiedowns are required per stack, not per trailer load.

- **CONTAINMENT WALL AND SIDEBOARD — WHAT KEEPS SCRAP IN**
  Structure extending to the full height of the load that blocks movement in the direction it faces. More walls mean fewer tiedowns required. Sideboards are removable boards serving the same purpose.

I CONTAINERS, BOXES, AND ROCK

- **CORNER FITTING AND TWISTLOCK — HOW A CONTAINER LOCKS DOWN**
  Corner fittings are the reinforced castings at the eight corners of a shipping container. A twistlock is the pin on the chassis that enters a fitting and turns to lock. These are what the half-inch movement tolerances are measured against.

- **INTEGRAL LOCKING DEVICE — SECUREMENT BUILT INTO THE VEHICLE**
  A locking mechanism forming part of the trailer rather than a separate tiedown. Where a vehicle has an **integral securement system**, the roll-off container rules in 393.134 do not apply to it.

- **ROLL-ON/ROLL-OFF AND HOOK LIFT — THE ROLL-OFF BOX**
  Open-topped containers loaded by being dragged or hooked onto a tilting truck bed. What most people call a dumpster or skip.

- **LIFTING DEVICE AND STOPS — THE MECHANISM AND THE BLOCKS**
  The hook or hoist arm that loads the container, and the fixed blocks it seats against. Both count as securement in their own right, which is why damaged ones require replacement tiedowns.

- **SIDE RAIL — THE STRUCTURAL EDGE OF THE TRUCK FRAME**
  The long member running down each side of the chassis, used as an attachment point for lengthwise tiedowns.

- **OVERHANG — LOAD EXTENDING PAST THE TRAILER**
  How far cargo projects beyond the front or rear of the deck. Capped at five feet for an empty container carried off-chassis.

- **CRIB — A TIMBER FRAME FIXED TO THE DECK**
  Hardwood built up on the deck so a rounded boulder rests on both deck and timber, with at least three separated contact points stopping it rolling.

- **HARDWOOD — SPECIFIED BY NAME FOR BOULDERS**
  Not any timber. The boulder section calls for hardwood blocking at least 4 by 4 inches in section, and a crib built of hardwood.

- **SHACKLE — A CONNECTING LINK**
  A U-shaped fitting closed with a pin, used to join two chains. Required where two chains cross over a non-cubic boulder.

J ON THE ROAD

- **ROADSIDE INSPECTION — A CHECK AT A SCALE OR BY THE ROADSIDE**
  An enforcement officer examines the vehicle, the driver's documents, and the load. It produces a written report, which the carrier must review, correct, and certify back to the issuing agency.

- **OUT OF SERVICE — THE VEHICLE STOPS HERE**
  A declaration that the vehicle may not be operated until the defect is corrected. Not a warning and not a citation — an order. Driving in violation of one carries penalties far beyond the original fault.

- **CVSA — THE COMMERCIAL VEHICLE SAFETY ALLIANCE**
  The body that publishes the North American Standard Out-of-Service Criteria, the annually revised document inspectors use to decide which violations put a vehicle out of service. Separate from the regulations themselves.

- **DUTY STATUS — WHAT THE DRIVER IS CURRENTLY DOING**
  Off duty, sleeper berth, driving, or on duty not driving. Changing between them is one of the three triggers for re-examining the load.

- **TAILBOARD AND TAILGATE — THE BACK EDGE OF THE DECK OR BODY**
  Part of the truck's own equipment that must be secured, alongside doors, tarpaulins, and the spare tire.

- **TARPAULIN — THE TARP OVER A FLATBED LOAD**
  Weather protection, not securement. It is itself equipment that must be secured, whether spread over a load or bundled on an empty deck.

- **SEALED LOAD — A TRAILER THE DRIVER MAY NOT OPEN**
  Closed and sealed by the shipper, with the driver instructed not to break the seal. Reported to affect the en-route inspection duty, though the exact wording is worth reading at 392.9(b).

K DIRECTIONS, AND READING A CITATION

DIRECTIONS

- **LONGITUDINAL**
  Front to back, along the vehicle.
- **TRANSVERSE AND LATERAL**
  Side to side, across the vehicle.
- **VERTICAL**
  Up and down — the direction the 20 percent downward force rule addresses.
- **TIER**
  A layer of cargo. A second tier sits on the first.

ABBREVIATIONS

- **CFR**
  Code of Federal Regulations.
- **FMCSA**
  Federal Motor Carrier Safety Administration.
- **CMV**
  Commercial motor vehicle.
- **WLL**
  Working load limit.
- **ASTM**
  The standards body whose specification governs steel strapping.

49 CFR 393.106(D)

PIECE	WHAT IT MEANS
49	**Title 49 — the transportation title**
CFR	**Code of Federal Regulations**

PIECE	WHAT IT MEANS
393	**Part 393 — parts and accessories necessary for safe operation**
.106	**Section 106 — general requirements for securing cargo**
(d)	**The paragraph inside it — here, the aggregate working load limit rule**

Every rule in this guide can be read free at **ecfr.gov**, the government's own current text. Not a summary site and not a vendor's page — the regulation itself. If anything you are told anywhere contradicts what you read there, the regulation wins.

THE RULES THAT APPLY TO EVERY LOAD

49 CFR Part 393, Subpart I. Cargo securement is the violation small carriers collect most often, and almost always because of arithmetic rather than carelessness. This part is the general system: the forces the law assumes, how a tiedown's rating is counted, and how many you need. Get this right and the commodity chapters become detail.

1.1 WHAT THE RULES COVER

393.100. The scope is broader than most drivers assume, and it is not limited to flatbeds.

- **APPLIES TO — TRUCKS, TRUCK TRACTORS, SEMITRAILERS, FULL TRAILERS, AND POLE TRAILERS**
  A dry van is not exempt. The rules for a sided vehicle are different from a flatbed, but they are still rules.
- **TWO SEPARATE DUTIES — AND BOTH ARE CITED SEPARATELY**
  393.100(b) requires the load be secured to prevent cargo leaking, spilling, blowing or falling from the vehicle. 393.100(c) separately requires cargo be contained, immobilized or secured to prevent shifting on or within the vehicle to an extent that affects stability or maneuverability.
- **BULK COMMODITIES ARE OUTSIDE THE GENERAL SECTION — NOT OUTSIDE THE LAW**
  393.106 excludes commodities in bulk that lack structure or fixed shape — liquids, gases, grain, liquid concrete, sand, gravel, aggregates — when carried in a tank, hopper, or box forming part of the vehicle structure.
- **COMMODITY RULES BEAT GENERAL RULES — WHERE THEY ADD REQUIREMENTS**
  393.116 through 393.136 cover logs, dressed lumber, metal coils, paper rolls, concrete pipe, intermodal containers, automobiles and light trucks, heavy equipment, crushed vehicles, roll-on containers, and large boulders. Where one of those applies, it takes precedence.

"IT DIDN'T FALL OFF" IS NOT THE STANDARD

Nothing in Subpart I asks whether the load arrived intact. The requirement is that the securement system be capable of withstanding specified forces, and an inspector evaluates the system in front of them — the count, the ratings, the condition, the edge protection. A load that made it through four states with two straps still fails on the roadside if it needed four.

1.2 THE FORCES THE LAW ASSUMES

393.102(a). Two sets of numbers, applied separately, and they are not the same set. This is the engineering behind every rule that follows.

MINIMUM PERFORMANCE CRITERIA

DIRECTION	AGAINST BREAKING STRENGTH	AGAINST WORKING LOAD LIMIT
Forward deceleration	**0.8 g**	**0.435 g**
Rearward acceleration	**0.5 g**	**0.5 g**
Lateral acceleration	**0.5 g**	**0.25 g**

Forward is the punishing direction — 0.8 g, well over half the weight of the load thrown at whatever is holding it. That is a hard stop, and it is why the front of a load is where the rules concentrate. Note also that the two columns differ: a system can be inside its breaking strength and still exceed its working load limit.

THE 20 PERCENT DOWNWARD RULE

393.102(b). If the article is not fully contained within the structure of the vehicle, the securement system must provide a downward force of at least 20 percent of the weight of the article. Contained inside the vehicle structure, it may instead be secured under the general rule in 393.106(b).

1.3 THREE WAYS TO COMPLY

393.102(c). Securement is considered to meet the performance requirements by any one of these. Knowing there are three is worth money, because two of them need no tiedowns at all.

IMMOBILISED

IT CANNOT SHIFT OR TIP

So that it cannot shift or tip to an extent that the vehicle's stability or maneuverability is adversely affected. Blocking, bracing, dunnage, or a tight load that fills the space.

SIDED VEHICLE

WALLS OF ADEQUATE STRENGTH

Each article in contact with, or sufficiently close to, a wall or other articles so it cannot shift or tip to that extent. This is how most van freight complies.

TIEDOWNS

PER 393.104 THROUGH 393.136

The route everyone thinks of first. Counted, rated, and arranged per the rules in the rest of this guide.

A TIGHT VAN LOAD IS A SECUREMENT SYSTEM

A trailer loaded wall to wall with no voids satisfies 393.102(c)(2) on its own. The compliance problem in vans is not straps — it is the partial load with a metre of empty space behind it and nothing filling the void.

1.4 THE FOUR RULES ON USING TIEDOWNS

393.104(f). Short, absolute, and responsible for a large share of roadside violations.

- **NO KNOTS — IN ANY TIEDOWN OR SECURING DEVICE**
  Not "avoid knots where possible". Tiedowns and securing devices must not contain knots.
- **REPAIRS FOLLOW THE STANDARD — OR THE MANUFACTURER'S INSTRUCTIONS**
  A field-repaired strap that meets neither is a damaged device, and damaged devices are prohibited outright by 393.104(b).
- **IT MUST NOT BE ABLE TO COME LOOSE — BY ITSELF**
  Each tiedown must be attached and secured in a manner that prevents it becoming loose, unfastening, opening or releasing in transit.

– **EDGE PROTECTION WHENEVER THERE IS ABRASION OR CUTTING — NOT JUST ON SHARP STEEL**
Required wherever a tiedown would be subject to abrasion or cutting at the point it touches the cargo, and it must resist abrasion, cutting **and crushing**. A folded piece of cardboard does not resist crushing.

ONE EXCEPTION WORTH KNOWING

393.128(b)(4): edge protectors are not required for synthetic webbing where the webbing contacts the tires of an automobile, light truck or van being transported. That is the only place the requirement is lifted.

THE DRIVER MUST BE ABLE TO TIGHTEN THEM

393.112 requires every tiedown, connector, or attachment mechanism to be designed, constructed and maintained so the driver of an in-transit vehicle can tighten it. A binder that no longer adjusts is a violation before anyone measures anything. Steel strapping is the sole exception.

1.5 THE 50 PERCENT RULE, AND HOW TIEDOWNS ARE COUNTED

393.106(d). The single most misunderstood calculation in trucking, because the same strap is worth different amounts depending on how it is run.

1 THE TARGET: AT LEAST HALF THE WEIGHT OF THE CARGO

The aggregate working load limit of the tiedowns securing an article or group of articles must be at least one-half times the weight of that article or group. A 40,000 lb load needs an aggregate working load limit of at least 20,000 lb.

2 VEHICLE ANCHOR TO AN ANCHOR POINT ON THE CARGO: COUNT HALF

A tiedown running from an anchor point on the vehicle to an anchor point on the article contributes one-half of its working load limit.

3

OVER THE LOAD AND BACK TO THE SAME SIDE: COUNT HALF

A tiedown attached to a vehicle anchor point, passing through, over or around the cargo, and returning to an anchor point on the

SAME SIDE

also contributes only one-half.

4

OVER THE LOAD TO THE OTHER SIDE: COUNT FULL

A tiedown running from an anchor point on the vehicle, through, over or around the article, and attaching to an anchor point

ON THE OTHER SIDE

contributes its full working load limit.

THIS IS WHERE COMPLIANT-LOOKING LOADS FAIL

Two carriers can run identical straps in identical numbers and one is legal while the other is not, purely because of where the ends are anchored. A strap thrown over the load and hooked back to the same rail is worth **half** what the same strap is worth crossing to the opposite rail. If you have been counting every strap at its full rating, you have been overestimating your securement by up to double.

AND THE RATING IS THE WEAKEST LINK

393.108(a): the working load limit of a tiedown is the lowest working load limit of any of its components — including the tensioner — or of the anchor point it attaches to, whichever is less. A 5,400 lb chain on a 3,000 lb binder is a 3,000 lb tiedown. Marked ratings on the device govern; where the manufacturer has not marked it, the tables in 393.108 apply.

1.6 HOW MANY TIEDOWNS

393.110. These counts are in addition to the 50 percent rule, not instead of it. You must satisfy both.

MINIMUM NUMBER OF TIEDOWNS

When the article is **not** blocked or positioned to prevent forward movement by a headerboard, bulkhead, other secured cargo, or other blocking — 393.110(b).

ARTICLE	MINIMUM TIEDOWNS
5 ft or less in length, and 1,100 lbs or less in weight	**1**
5 ft or less in length, but more than 1,100 lbs	**2**
Over 5 ft and up to 10 ft, any weight	**2**
Over 10 ft	**2, plus 1 more for every additional 10 ft or fraction**

If the article IS blocked against forward movement — by a headerboard, bulkhead, adequately secured cargo, or appropriate blocking — 393.110(c) applies instead, and the requirement drops to at least one tiedown for every 10 feet of length or fraction thereof. Blocking the front of the load is not decoration; it halves the strap count on long articles.

WORK A REAL EXAMPLE BEFORE YOU LOAD

A 16 ft article, unblocked: two tiedowns for the first 10 feet, plus one more for the remaining 6 feet as a fraction of 10 — three minimum. Then check the arithmetic separately: three straps must also total at least half the article's weight in aggregate working load limit, counted by the rules in 1.5. Whichever demands more, wins.

THE SPECIAL-PURPOSE EXCEPTION

393.110(d) lifts the counting rules for a vehicle carrying articles — machinery, structural steel, crane booms, girders, trusses — which because of design, size, shape or weight must be fastened by special methods. The load must still be securely and adequately fastened. This is a narrow exception for genuinely unusual freight, not a general escape for anything awkward.

1.7 PLACEMENT, ROLLING, AND THE FRONT END

ANYTHING THAT COULD ROLL MUST BE RESTRAINED — 393.106(C)(1)

By chocks, wedges, a cradle or equivalent, and the restraint must not be capable of becoming unintentionally unfastened or loose in transit.

- **ARTICLES SIDE BY SIDE UNDER TRANSVERSE TIEDOWNS — TOUCH, OR BE BLOCKED**
  393.106(c)(2): they must either be in direct contact with each other, or be prevented from shifting towards each other in transit.

- **FRONT END STRUCTURE HEIGHT — 4 FEET, OR HIGH ENOUGH TO BLOCK THE CARGO**
  393.114(b): it must extend to 4 feet above the floor, or to the height at which it blocks forward movement of the cargo, whichever is lower. Its width must equal the vehicle's width or block the cargo, whichever is narrower.

- **FRONT END STRUCTURE STRENGTH — HALF THE CARGO WEIGHT, OR FOUR-TENTHS IF TALL**
  393.114(c): under 6 feet in height, it must withstand a horizontal forward static load equal to half the weight of the cargo. At 6 feet or higher, four-tenths of that weight, distributed over the whole structure.

- **AND IT MUST RESIST PENETRATION — WITH NO GAPS**
  393.114(d): capable of resisting penetration by cargo contacting it at a deceleration of 20 feet per second per second, with no aperture large enough for an article in contact with it to pass through.

1.8 BEFORE YOU MOVE ON

Confirmed whether a commodity-specific rule applies to what you haul, or only the general rules

Checked that every tiedown and binder in your kit carries a legible manufacturer's working load limit marking

Identified the weakest component in each assembly — including the tensioner and the anchor point

Learned to count same-side tiedowns at half and cross-side tiedowns at full

Worked the 50 percent arithmetic on the heaviest load you actually haul

Worked the minimum tiedown count on your longest article, blocked and unblocked

Bought edge protection that resists crushing, not only abrasion

Checked that every binder still adjusts, and replaced the ones that do not

Inspected straps and chains for cuts, cracks, and weakened components, and pulled the damaged ones from service

WORKING LOAD LIMITS
WHAT YOUR GEAR IS ACTUALLY RATED FOR

Part One gave you the arithmetic. This part gives you the numbers to put into it. Every figure here is from the tables printed in 49 CFR 393.108 — and the most valuable pages are the ones telling you what the law assumes when your equipment has no marking on it at all.

2.1 WHERE A RATING COMES FROM

393.108(a) and (b). Two sources, and one of them beats the other.

- **THE MANUFACTURER'S MARKING WINS — EVEN WHEN IT DIFFERS FROM THE TABLES**
  Tiedown material marked by the manufacturer with a working load limit that differs from the table value is considered to have the working load limit it is marked with. The tables are the fallback, used when the material is not marked.

- **THE ASSEMBLY IS ONLY AS STRONG AS ITS WEAKEST PART — TENSIONER INCLUDED**
  The working load limit of a tiedown, connector or attachment mechanism is the lowest working load limit of any of its components, **including the tensioner**, or the working load limit of the anchor point it attaches to, whichever is less.

- **WORKING LOAD LIMIT IS NOT BREAKING STRENGTH — AND THE DIFFERENCE IS DELIBERATE**
  Breaking strength is where it fails. Working load limit is what it may be worked at. Every calculation in Subpart I runs on working load limit, and a device advertised by its breaking strength is telling you the number that does not count.

THE BINDER IS USUALLY THE WEAK LINK

Operators buy chain by grade and then attach it to whatever binder was in the truck. A 3/8 inch Grade 70 chain rated 6,600 lb running through a binder rated 5,000 lb is a **5,000 lb tiedown**, and that is the number that goes into your total. Buy the binder to match the chain, and check the anchor point rating too — it is in the same sentence of the regulation for a reason.

2.2 WHAT THE LAW ASSUMES WHEN NOTHING IS MARKED

This is the section that costs people money. An unmarked device is not excluded from your total — it is counted at the lowest plausible value, which is almost always far less than what you paid for.

DEFAULT RATINGS FOR UNMARKED EQUIPMENT

UNMARKED ITEM	TREATED AS	CITE
Welded steel chain, no grade or rating	**Grade 30 proof coil — the lowest grade in the table**	**393.108(d)**
Wire rope, no working load limit marked	**One-fourth of nominal strength from the Wire Rope Users Manual**	**393.108(e)(1**
Wire rope, construction type not identifiable	**6 × 37 fiber core**	**393.108(e)(2**
Synthetic cordage, composition not identifiable	**Polypropylene fiber rope — the weakest of the synthetics**	**393.108(c)**
Manila rope, no marking	**Rated by diameter from the table**	**393.108(f)**

UNMARKED ITEM	TREATED AS	CITE
Steel strapping, no working load limit marked	**One-fourth of breaking strength in ASTM D3953-97**	**393.104(e) n**
Friction mat, unmarked or unrated	**Resistance equal to 50% of the weight placed on it**	**393.108(g)**

Look at what the chain default does to you. A 3/8 inch Grade 70 chain is rated 6,600 lb. Rub the grade markings off, or buy chain that never had them, and the same chain is counted at **2,650 lb** — the Grade 30 value. You lose 60 percent of your securement on paper without losing an ounce of steel in reality.

READING CHAIN GRADE MARKS

Grade is stamped on the links, and it appears in one of three forms. Grade 30 shows as **3**, **30**, or **300**. Grade 43 as 4, 43 or 430. Grade 70 as 7, 70 or 700. Grade 80 as 8, 80 or 800. Grade 100 as 10, 100 or 1000. Transport chain is usually Grade 70 and often gold-coloured, but the colour is a convention, not a rating — read the stamp.

2.3 CHAIN

WORKING LOAD LIMITS, CHAIN (POUNDS)

393.108. Grade 80 and 100 are alloy; note that at 5/16 inch, Grade 80 is rated *lower* than Grade 70.

SIZE	GR 30	GR 43	GR 70	GR 80	GR 100
1/4"	**1,300**	**2,600**	**3,150**	**3,500**	**4,300**
5/16"	**1,900**	**3,900**	**4,700**	**4,500**	**5,700**
3/8"	**2,650**	**5,400**	**6,600**	**7,100**	**8,800**
7/16"	**3,700**	**7,200**	**8,750**	**—**	**—**
1/2"	**4,500**	**9,200**	**11,300**	**12,000**	**15,000**

SIZE	GR 30	GR 43	GR 70	GR 80	GR 100
5/8"	**6,900**	**13,000**	**15,800**	**18,100**	**22,600**

Grade 70 is the column most flatbed operators live in. Two 3/8 inch Grade 70 chains run over a load and anchored to the opposite side count at their full 6,600 lb each — 13,200 lb aggregate, enough for a load up to 26,400 lb on the 50 percent rule. Anchor those same two chains back to the same side and you have 6,600 lb, good for only 13,200 lb of cargo.

2.4 WEBBING, WIRE ROPE, AND STEEL STRAPPING

SYNTHETIC WEBBING

- **1-3/4 INCH**
  1,750 lb
- **2 INCH**
  2,000 lb
- **3 INCH**
  3,000 lb
- **4 INCH**
  4,000 lb

STEEL STRAPPING

- **1-1/4 × 0.029, 0.031, 0.035**
  1,190 lb
- **1-1/4 × 0.044 AND 0.050**
  1,690 lb
- **1-1/4 × 0.057**
  1,925 lb
- **2 × 0.044 AND 0.050**
  2,650 lb

WIRE ROPE, 6 × 37 FIBER CORE (POUNDS)

DIAMETER	WLL	DIAMETER	WLL
1/4"	**1,400**	5/8"	**8,300**
5/16"	**2,100**	3/4"	**10,900**
3/8"	**3,000**	7/8"	**16,100**
7/16"	**4,100**	1"	**20,900**
1/2"	**5,300**		

Two rules specific to steel strapping, from the notes to 393.104(e). Strapping **1 inch or wider must have at least two pairs of crimps in each seal**, and where an end-over-end lap joint is formed it must be sealed with **at least two seals**. Strapping is also the single exception to the rule in 393.112 that the driver must be able to tighten a tiedown in transit.

2.5 ROPE

Four synthetics and manila, all rated by diameter. The spread between them is large, which is why an unidentifiable rope defaults to the weakest.

WORKING LOAD LIMITS, ROPE (POUNDS)

Three-strand and eight-strand constructions. Manila per 393.108(f).

DIAMETER	MANILA	POLYPROPYLENE	POLYESTER	NYLON	DOUBLE BRAID NYLON
3/8"	**205**	**400**	**555**	**278**	**336**
7/16"	**265**	**525**	**750**	**410**	**502**
1/2"	**315**	**625**	**960**	**525**	**655**
5/8"	**465**	**925**	**1,500**	**935**	**1,130**
3/4"	**640**	**1,275**	**1,880**	**1,420**	**1,840**
1"	**1,050**	**2,100**	**3,300**	**2,520**	**3,250**

Rope is weak, and manila is weakest of all. A 1 inch manila rope contributes 1,050 lb — less than a 2 inch strap. Rope is rarely a practical primary tiedown for heavy freight, and where it is used, the numbers have to be counted honestly rather than assumed to be "about the same" as a strap.

LABEL YOUR ROPE, OR LOSE TWO-THIRDS OF IT

Polyester at 1 inch is rated 3,300 lb. If nobody can tell from looking at it what it is made of, 393.108(c) says it counts as polypropylene — **2,100 lb**. Same rope, same strength, 36 percent less credit, because it could not be identified at the roadside.

2.6 TWO WORKED EXAMPLES

1

A 14,000 LB PIECE OF EQUIPMENT, UNBLOCKED, 12 FT LONG

COUNT CHECK:
over 10 ft, so two tiedowns plus one more for the extra 2 ft as a fraction of 10 — three minimum.

RATING CHECK:
aggregate working load limit must be at least 7,000 lb. Three 3/8 inch Grade 70 chains crossed to the opposite side count full: 3 × 6,600 = 19,800 lb. Comfortable.

BUT
if those same chains run over the load and back to the same rail, they count at half: 3 × 3,300 = 9,900 lb. Still passes, with far less margin than it looks.

2

A 9,000 LB LOAD ON 2 INCH STRAPS

RATING CHECK:
aggregate must be at least 4,500 lb. A 2 inch strap is 2,000 lb. Crossed to the opposite side, three straps give 6,000 lb and it passes.

SAME-SIDE:
those three straps count at 1,000 lb each — 3,000 lb aggregate, and the load is

ILLEGAL
despite looking identical from the roadside.

AND CHECK THE RATCHET:
if the hardware is rated below 2,000 lb, the strap is worth the hardware's number, not the webbing's.

DO THIS ONCE FOR YOUR OWN EQUIPMENT

Write down the rated working load limit of every chain, strap, binder and ratchet you own, then work out the heaviest load each combination can legally secure. It is an hour of arithmetic done once, and it turns a roadside inspection from a question into a known answer.

2.7 BEFORE YOU MOVE ON

Checked every chain in the truck for a legible grade stamp, and retired or relabelled the unmarked ones

Checked that binders and ratchets carry their own rated markings

Identified the lowest-rated component in each assembly you actually use

Confirmed the anchor points on your trailer are rated, and know that number

Confirmed your rope is identifiable by composition, if you use rope at all

Checked steel strapping seals: two pairs of crimps per seal at 1 inch or wider, two seals at a lap joint

Written down the aggregate working load limit your normal kit provides, crossed and same-side

Compared that figure against the heaviest load you regularly haul

METAL COILS AND PAPER ROLLS

Two commodities, one shared problem: both are heavy, both are round, and both kill people when they get loose. The regulations respond with the most exacting rules in Subpart I — including several things that are flatly prohibited no matter how secure they look.

3.1 WHEN THESE RULES APPLY

Both sections turn on the same weight figure, and both let lighter shipments fall back to the general rules.

393.120

METAL COILS

Applies to one or more metal coils which, individually or grouped together, weigh **5,000 lb or more**. Lighter shipments may be secured under the general rules in 393.100 through 393.114.

393.122

PAPER ROLLS

Applies to shipments of paper rolls which, individually or together, weigh **5,000 lb or more**. Lighter shipments, and rolls unitized on a pallet, may use either these rules or the general ones.

ORIENTATION

DECIDES EVERYTHING

Both sections branch on which way the eye of the coil or roll is pointing — vertical, crosswise, or lengthwise. Identify that first; the rest follows from it.

"EYES" MEANS THE HOLE THROUGH THE MIDDLE

Eyes vertical — the hole points up at the sky, the coil sits like a doughnut on a table. **Eyes crosswise** — the hole points out at the sides of the trailer, so the coil would roll forwards and backwards. **Eyes lengthwise** — the hole points forward, so it would roll side to side. Every rule below depends on getting this right.

3.2 METAL COILS, EYES VERTICAL

393.120(b). On a flatbed, in a sided vehicle, or in an intermodal container with anchor points.

1 ONE TIEDOWN DIAGONALLY, LEFT FRONT TO RIGHT REAR

Attached from the left side of the vehicle near the forwardmost part of the coil, across the eye, to the right side near the rearmost part.

2 ONE TIEDOWN DIAGONALLY THE OTHER WAY

From the right side near the forwardmost part, across the eye, to the left side near the rearmost part. The two diagonals together are what stop it tipping.

3 ONE TIEDOWN TRANSVERSELY OVER THE EYE

Straight across the top.

4 SOMETHING STOPPING FORWARD MOVEMENT

Blocking and bracing, friction mats, or tiedowns to prevent longitudinal movement in the forward direction.

COILS GROUPED IN ROWS

Where coils are loaded side by side in a transverse or longitudinal row, each row needs: at least one tiedown at the front restraining forward motion; at least one at the rear restraining rearward motion — both at **no more than 45 degrees** to the floor when viewed from the side, wherever practicable; at least one tiedown over the top of each coil or transverse row, positioned as close as practicable to the eye; and the whole arrangement must prevent shifting or tipping in every direction.

3.3 METAL COILS, EYES CROSSWISE AND LENGTHWISE

- **STOP IT ROLLING FIRST — AND HOLD THE CHOCKS WITH MORE THAN HOPE**
  Timbers, chocks, wedges or a cradle, which must **support the coil off the deck** and must not be capable of becoming unintentionally unfastened or loose in transit. If timbers, chocks or wedges are used they must be held in place by coil bunks or similar devices.

- **NAILED BLOCKING IS PROHIBITED AS THE SOLE MEANS — AND SO IS A NAILED WOOD CRADLE**
  Using nailed blocking or cleats as the only thing securing timbers, chocks or wedges is not permitted. Neither is a nailed wood cradle. Nails are not a securement system.

- **EYES CROSSWISE — ONE TIEDOWN THROUGH THE EYE EACH WAY**
  At least one through the eye restricting forward motion and at least one through the eye restricting rearward motion, each at no more than 45 degrees to the floor when viewed from the side, wherever practicable.

- **EYES LENGTHWISE — THREE PERMITTED OPTIONS**
  Two diagonals through the eye plus a transverse over the top plus blocking or friction mats; or two straight-through tiedowns each returning to the same side plus a transverse plus blocking or friction mats; or simply one tiedown over the top near the front, one over the top near the rear, and blocking or friction mats.

THE X-PATTERN PROHIBITION

With coils **eyes crosswise**, attaching tiedowns diagonally through the eye so they form an X when viewed from above the vehicle is **prohibited outright** — 393.120(c)(2). This is not a preference and there is no equivalent-means argument for it. It looks tidy, it looks strong, and it is a violation.

WITHOUT ANCHOR POINTS, THE RULES CHANGE SHAPE

Coils carried in a sided vehicle or intermodal container that has **no anchor points** must simply be loaded to prevent shifting and tipping, and may be secured using blocking and bracing, friction mats, tiedowns, or a combination. The obligation does not disappear; the method becomes yours to choose.

3.4 PAPER ROLLS: THE TIPPING RATIOS

393.122(b)(4). A tall narrow roll tips; a squat wide one does not. The regulation draws that line with three specific ratios, and nobody remembers them without a table.

WHEN A PAPER ROLL MUST BE PREVENTED FROM TIPPING

Eyes vertical, in a sided vehicle, where the roll is not already held by vehicle structure or other cargo.

SITUATION	RATIO	REQUIREMENT
Any roll, tipping sideways or rearwards	**Width > 2 × diameter**	**Band to other rolls, brace, or tie down**
Forwardmost roll in a group	**Width > 1.75 × diameter**	**Band, brace, or tie down**
Forwardmost roll, restrained forward by friction mats alone	**Width ≤ 1.75 × diameter**	**No banding, bracing or tiedowns needed**
Any roll or forwardmost roll, not restrained by friction mats alone	**Width > 1.25 × diameter**	**Band, brace, or tie down**

Read the third and fourth rows together. A friction mat doing the forward restraint on its own buys you a more generous ratio — 1.75 instead of 1.25. That is a real operational difference, and it is the sort of detail that decides whether a load needs banding at all.

FRICTION MATS HAVE TO POINT THE RIGHT WAY

393.122(b)(6): a friction mat used as the principal securement for a paper roll **must protrude from beneath the roll in the direction it is providing securement.** A mat tucked entirely under the roll is not doing the job the regulation credits it for.

3.5 PAPER ROLLS: LOADING AND STACKING

EYES VERTICAL, ON THE FLOOR

- **TIGHT AGAINST WALLS, ROLLS, OR OTHER CARGO**
  That contact is the primary securement.
- **NOT ENOUGH ROLLS TO REACH THE WALLS**
  Prevent lateral movement by filling the void, blocking, bracing, tiedowns or friction mats. The rolls may also be banded together.
- **A VOID BEHIND THEM BIGGER THAN A ROLL DIAMETER**
  Prevent rearward movement by friction mats, blocking, bracing, tiedowns, or banding to other rolls.
- **IF BANDED**
  Rolls tight against each other forming a stable group, bands applied tightly and secured so they cannot fall off the rolls or onto the deck.

EYES CROSSWISE, ON THE FLOOR

- **STOP IT ROLLING LENGTHWISE**
  By contact with structure or cargo, by chocks, wedges or blocking of adequate size, or by tiedowns.
- **CHOCKS HELD BY MORE THAN FRICTION**
  Secured by some additional means so they cannot work loose in transit.
- **MORE THAN 8 INCHES TO THE WALL**
  Void fillers, blocking, bracing, friction mats or tiedowns to stop it shifting sideways.
- **NEVER USE THE DOORS**
  The rearmost roll must not be secured using the rear doors, or by blocking held in place by those doors.

STACKING HAS ITS OWN RULES, AND THEY ARE STRICT

Eyes vertical: nothing goes on an upper layer unless the layer beneath extends to the front of the vehicle. A blocking roll must be at least **1.5 inches taller** than the others, or raised at least 1.5 inches on dunnage — and a roll in the rearmost row of any layer raised on dunnage may **not** be secured by friction mats alone.

Eyes crosswise: no second layer unless the bottom extends to the front, and no third or higher layer unless every well in the layer beneath is filled. The foremost roll in each upper layer, or any roll with an empty well in front of it, must be banded, or blocked against an eye-vertical blocking roll at least 1.5 times taller than the roll it blocks, or sat in a well formed by two lower rolls of equal or greater diameter.

Eyes lengthwise: no roll goes in a higher layer if another roll would still fit in the layer beneath.

3.6 PAPER ROLLS ON A FLATBED

393.122(i). Curtain-sided counts as a flatbed here — fabric sides are weather protection, not walls.

- **LOAD IT AS THOUGH IT WERE A SIDED VEHICLE — THEN ADD TIEDOWNS ON TOP**
  For eyes vertical or eyes lengthwise, the rolls must be loaded and secured as described for a sided vehicle, **and** the entire load secured by tiedowns meeting the general requirements of 393.100 through 393.114.
- **STACKED LOADS WITH EYES VERTICAL ARE PROHIBITED — ON A FLATBED**
  393.122(i)(1)(ii) states it flatly. There is no securement arrangement that makes this permissible.
- **EYES CROSSWISE — ROLLING FIRST, THEN LATERAL TIEDOWNS**
  Prevented from rolling or shifting longitudinally by contact, chocks, wedges, blocking or bracing of adequate size, or tiedowns; chocks held securely by more than friction; and tiedowns used per the general rules to prevent lateral movement.

> THE PATTERN ACROSS BOTH COMMODITIES
>
> Every rule in this part is one of three things: stop it rolling, stop it tipping, or stop it sliding forward. When you meet a situation the text does not obviously cover, ask which of the three is unaddressed — that is nearly always where the requirement is pointing.

3.7 BEFORE YOU MOVE ON

- Confirmed whether your shipment reaches 5,000 lb, individually or grouped
- Identified the orientation of the eyes before planning any securement
- Carry coil bunks or equivalent, so chocks and timbers are not held by nails
- Know that the X-pattern through the eye is prohibited on eyes-crosswise coils
- Measured width against diameter on paper rolls, to know which tipping ratio applies
- Position friction mats so they protrude in the direction they are restraining

Never plan to secure a rearmost roll against the trailer doors

Know that stacked eyes-vertical paper rolls are prohibited on a flatbed

Checked that upper-layer rules are satisfied before loading a second tier

LOGS, LUMBER, AND CONCRETE PIPE

Three commodities that share a shape problem: they are long, they are heavy, and left alone they roll. Logs get the most generous tiedown arithmetic in Subpart I and the strictest requirements about the vehicle itself. Lumber gets four permitted stacking methods. Concrete pipe gets a blocking specification measured in inches.

4.1 LOGS: WHEN THE LOG RULES APPLY

393.116(a). Three ways out of this section, and each sends you somewhere specific.

- **UNITIZED BY BANDING — USE THE GENERAL RULES INSTEAD**
  Logs banded or comparably bound into a single article may be transported under 393.100 through 393.114.
- **FOUR PROCESSED LOGS OR FEWER — GENERAL RULES**
  A load consisting of no more than four processed logs falls outside the log-specific section.
- **FIREWOOD, STUMPS, AND DEBRIS — MUST BE ENCLOSED**
  Firewood, stumps, log debris and other short logs must be carried in a vehicle or container **enclosed on both sides, front, and rear**, of adequate strength to contain them. Longer logs may be loaded that way too.
- **EVERYTHING ELSE — NEEDS A VEHICLE BUILT FOR IT**
  Logs must be transported on a vehicle designed, built or adapted for logs, fitted with bunks, bolsters, stakes or standards that **cradle the logs and prevent them rolling**. This is a requirement about the truck, not just the straps.

THE ONE-SIXTH RULE

393.116(b)(4): the aggregate working load limit for tiedowns securing a stack of logs on a frame vehicle, or a flatbed fitted with bunks, bolsters or stakes, must be at least **one-sixth the weight of the stack** — not the one-half that applies to general cargo.

That is not leniency. It reflects that the bunks and stakes are doing most of the work, and it only applies because they are there. Take the cradling structure away and you are back under the general rules.

4.2 LOGS: HOW THE LOAD MUST SIT

393.116(c). Most of log securement is about stacking, not strapping.

1 SOLIDLY PACKED, RESTING ON THE STRUCTURE

Logs must be solidly packed, and the outer bottom logs must be in contact with and resting solidly against the bunks, bolsters, stakes or standards.

2 EVERY OUTSIDE LOG TOUCHES AT LEAST TWO SUPPORTS

Each outside log on the side of a stack must touch at least two stakes, bunks, bolsters or standards. If one end does not actually touch, it must rest stably on other logs and extend beyond the support.

3 NOTHING RIDES ABOVE THE STAKES

The centre of the highest outside log on each side or end must be below the top of each stake, bunk or standard.

4 ANY LOG NOT HELD BY CONTACT GETS ITS OWN TIEDOWN

And additional tiedowns are required where the condition of the wood gives such low friction between logs that they are likely to slip on each other.

NUMBERS THAT CATCH PEOPLE ON CROSSWISE SHORTWOOD

The end of a log in the lower tier may not extend more than **one-third of the log's total length** beyond the nearest supporting structure. A single crosswise stack needs at least two tiedowns attached to the vehicle frame front and rear; where two are used they sit at approximately **one-third and two-thirds** of the log length. A vehicle more than **33 feet** long needs centre stakes dividing it into roughly equal sections.

Two stacks side by side: no space between them, the outside of each stack raised at least **1 inch** within 4 inches of the log ends or the vehicle side, the highest log no more than **8 feet** above the deck, and at least one tiedown lengthwise across each stack.

LENGTHWISE LOADS, AND POLE TRAILERS

Shortwood lengthwise: cradled in a bunk unit or contained by stakes, then either two tiedowns, or one tiedown positioned about midway between the stakes if the stack is blocked front and rear by structure or other stacks, or bound by at least two wrapper-type devices such as wire rope encircling the whole load. Wrappers used to bundle logs together need not be attached to the vehicle.

Longwood: cradled in two or more bunks, then either two tiedowns at effective locations or two wrappers. **Pole trailers:** one tiedown at each bunk, or two wrappers encircling the load with the front and rear wrappers at least **10 feet apart**.

4.3 DRESSED LUMBER AND BUILDING PRODUCTS

393.118. Bundles of dressed or packaged lumber, plywood, gypsum board and similar. Loose, unbundled lumber is not covered here — it is treated as loose items under the general rules.

- **BUNDLES GO SIDE BY SIDE — TOUCHING, OR BLOCKED**
  Placed in direct contact with each other, or a means provided to prevent them shifting towards each other.
- **ONE TIER — NOTHING SPECIAL**
  Bundles carried on a single tier are secured under the general provisions of 393.100 through 393.114.
- **A SIDED VEHICLE OF ADEQUATE STRENGTH — ALSO JUST THE GENERAL RULES**
  Loaded in a sided vehicle or container of adequate strength, lumber may be secured under the general provisions.

MORE THAN ONE TIER — FOUR PERMITTED METHODS, PICK ONE

This is where the section does its work, and the options are genuinely different from each other.

SECURING LUMBER IN MORE THAN ONE TIER

393.118(d). Any one of these four satisfies the rule.

METHOD	WHAT IT REQUIRES
Stakes	**Blocked against lateral movement by stakes on the sides of the vehicle, and se**
Blocking or friction between tiers	**Restrained laterally by blocking or high-friction devices between tiers, and s**
Spacers	**Placed directly on other bundles or on spacers, with tiedowns over the top tie**
Tiedowns on every tier	**Tiedowns over each tier, minimum two over each top bundle longer than 5 ft, ir**

The spacer method has extra conditions. A spacer must be long enough to support all pieces in the bottom row of the bundle, at least as wide as it is high, and if built from layers, those layers must be fastened so the spacer behaves as one piece. With three tiers, the middle and top bundles both need tiedowns. With more than three tiers, one middle bundle and the top bundle need them, and the middle tier being secured may not sit more than **6 feet above the deck**.

4.4 CONCRETE PIPE: BLOCKING AND ARRANGEMENT

393.124. Flatbeds and lowboys. Pipe bundled into a single rigid article with no tendency to roll, or carried in a sided vehicle, goes back to the general rules.

TIEDOWNS

- **AGGREGATE AT LEAST HALF THE GROUP WEIGHT**
  Not less than half the total weight of all the pipes in the group.
- **A TRANSVERSE TIEDOWN WORKS DOWNWARD**

BLOCKING

- **SYMMETRICAL ABOUT THE CENTRE**
  One or more pieces, placed symmetrically about the centre of a pipe.
- **THE PATTERN**

One through a pipe on an upper tier, or over the longitudinal tiedowns, is considered to secure all the pipes beneath on which it causes pressure.

- **GROUPS BY DIAMETER**
  Pipe of more than one diameter must be formed into groups of a single size, each secured separately.

One piece extending at least half the distance from centre to each end, and two pieces on the opposite side, one at each end.

- **TIMBER MINIMUM 4 × 6 INCHES**
  Placed firmly against the pipe and secured so it cannot move out from under it.

BELL PIPE HAS ITS OWN LOADING PATTERN

Bell pipe must sit on at least two longitudinal spacers tall enough to keep the bell clear of the deck. In a single tier the bells alternate on opposite sides of the vehicle. In more than one tier, the bells of the bottom tier all go on the same side, and every tier above has its bells on the opposite side to the tier below. Where a second tier is incomplete, bottom-tier pipes not supporting a pipe above go back to alternating.

4.5 CONCRETE PIPE: THE 45-INCH SPLIT

Inside diameter decides which set of tiedown rules applies. Both use the same 45-degree angle limit.

UP TO 45"

INSIDE DIAMETER

Bottom tier immobilised longitudinally at each end by blocking, vehicle end structure, stakes, a locked pipe unloader or equivalent. Every pipe in the bottom tier held firmly against the adjacent pipe by tiedowns through the front and rear pipes — at least one through the front pipe running aft at no more than 45 degrees, and at least one through the rear pipe running forward at no more than 45 degrees.

OVER 45"

INSIDE DIAMETER

Front and rear pipe immobilised by blocking, wedges, end structure, stakes or a locked pipe unloader. Then **every pipe** secured by a tiedown through the pipe: those in the front half, including the middle one if the count is odd, running aft at no more than 45 degrees; those in the rear half running forward at the same limit.

STACKING PIPE

The bottom tier covers the full length of the vehicle, or forms a partial tier in one or two groups. Pipe in an upper tier goes **only in the wells** formed by adjacent pipes in the tier beneath, and a third or higher tier may not be started unless every well in the tier below is filled.

TWO DETAILS THAT DECIDE A ROADSIDE CALL

Under 45 inches, if you are not tying each pipe individually, you need either one 1/2-inch chain or wire rope, or two 3/8-inch, laid longitudinally over the group — plus one transverse tiedown for every 10 feet of load length. And if the first or last pipe of a top-tier group is not sitting in the first or last well beneath it, that pipe needs an additional tiedown running back or forward at no more than 45 degrees.

Over 45 inches, if the front or rear pipe is not in contact with end structure, stakes or a locked pipe unloader, that pipe needs **two** tiedowns rather than one.

4.6 BEFORE YOU MOVE ON

- Confirmed whether your log load falls under the log rules at all, or under the general ones
- Checked that your log vehicle actually has bunks, bolsters or stakes that cradle the load
- Confirmed non-permanent stakes are secured against separating from the vehicle
- Worked the one-sixth arithmetic on your heaviest log stack
- Checked that no log rides with its centre above the top of a stake
- Chosen one of the four lumber tiering methods and know which conditions come with it
- Confirmed spacers are at least as wide as they are high, and long enough to support the bottom row
- Carry timber blocking of at least 4 by 6 inches for concrete pipe
- Know which side of the 45-inch inside diameter split your pipe falls on
- Checked that every upper-tier pipe sits in a well, with all wells beneath filled

VEHICLES, EQUIPMENT, AND CRUSHED CARS

Hauling something that has its own wheels changes the problem. It can roll, it can steer, it can articulate, and parts of it can move on their own. Three sections cover it, split at 10,000 pounds — and the third has the only outright ban on a material anywhere in Subpart I.

5.1 WHICH SECTION APPLIES

One number decides it, and it is the weight of the individual item — not the load.

393.128

10,000 LB OR LESS

Automobiles, light trucks and vans that individually weigh 4,536 kg (10,000 lb) or less. Minimum **two** tiedowns.

393.130

10,000 LB OR MORE

Heavy vehicles, equipment and machinery that operate on wheels or tracks — front end loaders, bulldozers, tractors, power shovels. Minimum **four** tiedowns.

393.132

FLATTENED OR CRUSHED

Automobiles, light trucks and vans that have been flattened or crushed. Its own rules entirely, regardless of weight.

BELOW 10,000 LB YOU HAVE A CHOICE

393.130(a) says equipment and machinery lighter than 10,000 lb **may** be secured under the heavy-equipment section, under 393.128, or under the general rules in 393.100 through 393.114. A small skid steer can legally be secured the heavy way. Where the item is awkward, the stricter method is often the easier one to defend.

5.2 CARS, LIGHT TRUCKS, AND VANS

393.128(b). Short section, four rules, and two of them are about where the tiedown attaches.

- **BOTH ENDS, ALL FOUR DIRECTIONS — MINIMUM TWO TIEDOWNS**
  Restrained at both the front and the rear to prevent lateral, forward, rearward and vertical movement, using a minimum of two tiedowns.
- **TIEDOWNS THAT ATTACH TO THE VEHICLE — MUST USE THE DESIGNED POINTS**
  Where a tiedown is designed to be affixed to the structure of the automobile, light truck or van, it must use the mounting points on that vehicle **specifically designed for that purpose**. Not a suspension arm, not a bumper, not whatever the hook reaches.
- **TIEDOWNS OVER THE WHEELS — MUST RESTRAIN IN THREE DIRECTIONS**
  A tiedown designed to fit over or around the wheels must provide restraint in the lateral, longitudinal and vertical directions. A strap that only presses down is not doing the job the section credits it for.
- **EDGE PROTECTION — THE ONE PLACE IT IS NOT REQUIRED**
  393.128(b)(4): edge protectors are not required for synthetic webbing at points where the webbing comes into contact with the tires. This is the only exception to the edge protection rule in all of Subpart I.

5.3 HEAVY EQUIPMENT AND MACHINERY

393.130. The section that covers most flatbed equipment hauling, and the preparation rules matter as much as the tiedowns.

1

LOWER AND SECURE EVERY ACCESSORY

Accessory equipment such as hydraulic shovels must be
COMPLETELY LOWERED AND SECURED TO THE VEHICLE
. A bucket resting in the air on hydraulics is not secured — hydraulics bleed down, and the load changes shape on the road.

2

STOP IT ARTICULATING

Articulated vehicles must be restrained in a manner that prevents articulation while in transit. A machine that can fold in the middle will do so under braking unless something stops it.

3

FOUR TIEDOWNS, MINIMUM

Equipment or machinery with crawler tracks or wheels must be restrained against movement in the lateral, forward, rearward and vertical directions using a minimum of four tiedowns.

4

AS CLOSE TO THE CORNERS AS PRACTICABLE

Each tiedown must be affixed as close as practicable to the front and rear of the vehicle, or to mounting points on it that have been specifically designed for that purpose.

FOUR TIEDOWNS IS THE FLOOR, NOT THE ANSWER

The four-tiedown minimum sits **alongside** the 50 percent aggregate working load limit rule, not instead of it. A 40,000 lb excavator needs aggregate working load limit of at least 20,000 lb. Four 3/8 inch Grade 70 chains crossed to the opposite side give 26,400 lb and pass. The same four chains anchored back to the same side give 13,200 lb — four tiedowns, and still illegal.

5.4 FLATTENED OR CRUSHED VEHICLES

393.132. Scrap hauling, and the rules exist because crushed cars shed parts and fluid across the road behind you.

SYNTHETIC WEBBING IS PROHIBITED

393.132(b). Using synthetic webbing to secure flattened or crushed vehicles is **prohibited**. It may only be used to connect wire rope or chain to anchor points on the truck — and even then, the webbing must not come into contact with the crushed cars at all, **regardless of whether edge protection is used**.

This is the only material banned outright anywhere in the securement rules. Crushed steel is all edges, and no amount of protection is accepted as sufficient.

FOUR PERMITTED ARRANGEMENTS

393.132(c). Containment walls reduce the tiedowns required, one for one.

CONTAINMENT	TIEDOWNS REQUIRED
Walls on all four sides, full height of the load, blocking forward, rearward and lateral movement	**None specified**
Walls on three sides, full height, blocking in the directions they cover	**2 per vehicle stack**
Walls on two sides, full height, blocking forward and rearward	**3 per vehicle stack**
No containment walls	**4 per vehicle stack**

Note that the count is **per vehicle stack**, not per load. Three stacks on an open trailer with no walls is twelve tiedowns, not four.

CONTAINING WHAT FALLS OUT

Where any of the tiedown arrangements above is used, the vehicle must also be equipped with a means to prevent **liquids leaking from the bottom** and **loose parts falling from the bottom and all four sides**, extending to the full height of the cargo. That containment may be structural walls, sides or sideboards, a suitable covering material, or a combination — and synthetic material **is** permitted for containing loose parts, even though it is banned as a tiedown here.

5.5 BEFORE YOU MOVE ON

Confirmed the individual weight of what you are hauling, since 10,000 lb decides the section

Located the designed tiedown points on the vehicles you commonly carry

Confirmed wheel straps restrain laterally and longitudinally, not just downward

Made lowering and securing accessory equipment part of your loading routine

Know how to prevent articulation on any articulated machine you haul

Checked the aggregate working load limit separately from the tiedown count

If hauling scrap: removed synthetic webbing from the securement plan entirely

If hauling scrap: counted tiedowns per stack rather than per load

If hauling scrap: confirmed the trailer contains fluids and loose parts on all four sides

CONTAINERS AND BOULDERS

The last three commodity sections, and the two furthest apart in character. Containers are engineered objects with corner fittings and tolerances measured in half-inches. Boulders are irregular rock, where the regulation has to describe in words what a machine would otherwise measure — and ends up specifying the timber, the contact points, and the only permitted tiedown material.

6.1 INTERMODAL CONTAINERS ON A CHASSIS

393.126(b). The tightest tolerances anywhere in Subpart I.

PERMITTED MOVEMENT

All lower corners must be secured with securement devices or integral locking devices that cannot unintentionally become unfastened in transit.

DIRECTION	MAXIMUM MOVEMENT
Forward	**1/2 inch**
Aft	**1/2 inch**
To the right	**1/2 inch**
To the left	**1/2 inch**
Vertically	**1 inch**

The front and rear must be secured independently. That phrase appears in every container paragraph, and it means exactly what it says: one arrangement holding both ends is not compliant, however strong it is. A worn or missing twistlock at one corner is a violation even with the other three locked.

WHAT'S INSIDE IS A SEPARATE QUESTION

393.126(a): cargo contained **within** an intermodal container is secured under the general rules in 393.100 through 393.114, or under the applicable commodity rules. Securing the container to the chassis says nothing about the freight inside it, and both are the carrier's responsibility.

6.2 CONTAINERS ON ANYTHING ELSE

393.126(c) and (d). Carrying a container on a flatbed rather than a purpose-built chassis, loaded or empty.

LOADED CONTAINER

- **ALL LOWER CORNERS REST ON THE VEHICLE**
  Or are supported by a structure capable of bearing the container's weight — and that support structure must itself be independently secured to the motor vehicle.
- **TWO PERMITTED ATTACHMENT PATTERNS**
  Chains, wire ropes or integral devices fixed to all lower corners; or crossed chains fixed to all upper corners.
- **FRONT AND REAR INDEPENDENTLY**
  Each chain, wire rope or locking device attached so it cannot be unintentionally unfastened in transit.

EMPTY CONTAINER

- **THE CORNER RULE RELAXES**
  Lower corners need not all rest on the vehicle or be supported, provided the four conditions beside are met.
- **STABLE BEFORE SECUREMENT**
  Balanced and positioned so the container is stable before any tiedowns are added.
- **OVERHANG CAPPED AT 5 FEET**
  On either the front or the rear of the trailer.
- **AND STILL FULLY SECURED**
  Must not interfere with maneuverability, and must be secured against lateral, longitudinal and vertical shifting.

6.3 ROLL-ON/ROLL-OFF AND HOOK LIFT CONTAINERS

393.134. Roll-off boxes and dumpsters. These rules apply where the vehicle is **not** equipped with an integral securement system.

1 BLOCKED AGAINST FORWARD MOVEMENT

By the lifting device, by stops, by a combination of both, or by another suitable restraint mechanism.

2 FRONT SECURED AGAINST LATERAL AND VERTICAL MOVEMENT

By the lifting device or another suitable restraint.

3 REAR SECURED BY ONE OF THREE METHODS

One tiedown attached to both the vehicle chassis and the container chassis; or two tiedowns installed lengthwise, each securing one side of the container to one of the vehicle's side rails; or two hooks or an equivalent mechanism securing both sides to the vehicle chassis at least as effectively.

4 REAR SECUREMENT WITHIN 6 FT 7 IN OF THE BACK

The mechanisms used to secure the rear end must be installed no more than two metres from the rear of the container.

BROKEN STOPS MEAN EXTRA TIEDOWNS, NOT A SHRUG

393.134(b)(5): where one or more of the front stops or lifting devices is **missing, damaged, or not compatible**, additional manually installed tiedowns must be used to secure the container, providing the same level of securement as the missing or damaged components. Incompatible equipment is named alongside broken equipment — a container that does not match the truck is treated the same as one whose stops have failed.

6.4 LARGE BOULDERS: GETTING IT TO SIT STILL

393.136. Applies to natural, irregularly shaped rock over **11,000 lb** or over 2 cubic metres, on an open vehicle or one whose sides are not rated to contain it.

- **BETWEEN 220 LB AND 11,000 LB — YOUR CHOICE OF SECTION**
  Rock in that range may be secured under this section or under the general rules — including rock inside a vehicle designed to carry it, or secured individually by tiedowns where each piece can be stabilised and adequately secured.

- **CUT OR FORMED ROCK WITH A STABLE BASE — ALSO EITHER ROUTE**
  Rock shaped to provide a stable base may be secured under this section or the general rules.

- **FLATTEST SIDE DOWN — AND NARROW END FORWARD**
  Each boulder placed with its flattest or largest side down. If it is tapered, the narrowest end must point towards the front of the vehicle.

- **HARDWOOD BLOCKING, AT LEAST 4 BY 4 INCHES — TWO PIECES MINIMUM**
  Supported on at least two pieces of hardwood blocking with side dimensions of at least 10 cm by 10 cm, extending the full width of the boulder, placed as symmetrically as possible and supporting at least three-quarters of its length.

- **IF IT COULD ROLL — BUILD A CRIB**
  Where the flattest side is rounded or partly rounded, the boulder goes in a crib of hardwood timber fixed to the deck, so it rests on both deck and timber with at least **three well-separated points of contact** preventing it rolling in any direction.

6.5 LARGE BOULDERS: CHAIN ONLY

ONLY CHAIN MAY BE USED

393.136(c)(1). No webbing, no wire rope, no synthetic anything. Where tiedowns are in direct contact with the boulder they should, where possible, sit in valleys or notches across the top, and they must be arranged to prevent sliding across the rock surface.

THREE SHAPES, THREE ARRANGEMENTS

SHAPE	ARRANGEMENT
Cubic	**At least two chain tiedowns placed transversely across the vehicle, as close a**
Non-cubic, stable base	**At least two chain tiedowns forming an X over the boulder, passing over its cer**

SHAPE	ARRANGEMENT
Non-cubic, unstable base	**One chain around the top, between one-half and two-thirds of its height, plus i**

Note what the unstable-base arrangement is doing. The chain around the boulder is not holding it down — it is creating something to attach to, because an irregular rock offers nothing else. The four chains then work against horizontal movement in every direction.

A CONTRADICTION WORTH NOTICING

Crossing tiedowns diagonally to form an X is **prohibited** through the eye of a coil carried eyes crosswise, and **required** over a non-cubic boulder with a stable base. Same pattern, opposite instruction. That is the clearest illustration in the whole guide that securement rules are commodity-specific rather than general good practice, and that reasoning by analogy from one commodity to another is how carriers get violations.

6.6 BEFORE YOU MOVE ON

Checked every twistlock and corner fitting before moving a container, not after

Confirmed front and rear are secured by independent arrangements

Remembered that securing the container says nothing about the freight inside it

If carrying an empty container off-chassis: kept overhang within five feet

Confirmed roll-off rear securement sits within 6 ft 7 in of the container's rear

Know that damaged or incompatible stops require additional manual tiedowns

Carry hardwood blocking of at least 4 by 4 inches if you haul rock

Carry chain, and only chain, for boulders

Identified whether each boulder is cubic, stable-base, or unstable-base before loading

ON THE ROAD AND AT THE SCALE

Everything so far has been about loading. This part is about the rest of the trip — the checks the law requires once you are moving, what happens when an officer looks at your load, and how to turn all of it into a routine you actually follow rather than a chapter you read once.

7.1 THREE THINGS BEFORE THE WHEELS TURN

49 CFR 392.9(a). A driver may not operate, and a carrier may not require or permit a driver to operate, unless all three are true.

1 THE CARGO IS DISTRIBUTED AND SECURED

Properly distributed and adequately secured as specified in 393.100 through 393.136 — every rule in this guide, applied to what is actually on the trailer.

2 THE TRUCK'S OWN EQUIPMENT IS SECURED TOO

Tailgate, tailboard, doors, tarpaulins, spare tire, other equipment used in its operation, and the means of fastening the cargo. A loose tarp or an unsecured spare is a securement violation in its own right, and it has nothing to do with the freight.

3

NOTHING OBSTRUCTS THE DRIVER

Cargo or any other object must not obscure the view ahead or to either side, interfere with free movement of the arms or legs, prevent ready access to emergency accessories, or prevent the free and ready exit of any person from the vehicle.

THE SECOND ONE IS THE SLEEPER

Most operators think of securement as being about the load. 392.9(a)(2) puts the **truck's own gear** on the same footing: tarps, tailboards, doors, the spare. An empty flatbed with a loose tarp bundle is carrying an unsecured article of cargo.

7.2 THE CHECKS YOU OWE ONCE YOU ARE MOVING

392.9(b). Securement is not a one-time act at the dock. Straps settle and chains slack off as soon as a load takes the road, which is exactly what these intervals exist to catch.

WHEN TO STOP AND RE-CHECK

WHEN	WHAT IS REQUIRED
Before driving	**Satisfy yourself that all three requirements in 392.9(a) are met**
Within the first 50 miles	**Inspect the cargo and securing devices, and make any adjustment needed — includ**
At each change of duty status	**Reexamine and adjust**

WHEN	WHAT IS REQUIRED
Every 3 hours driven	**Reexamine and adjust**
Every 150 miles driven	**Reexamine and adjust**

The last three run on **whichever occurs first**. In practice that means a stop roughly every two to three hours on a long run, and it lines up naturally with fuel stops and the 30-minute break required under the hours of service rules. Plan them together and the requirement costs you almost nothing.

SEALED AND IMPRACTICABLE LOADS

The en-route inspection rules in 392.9(b) are widely reported not to apply to the driver of a **sealed** commercial motor vehicle who has been ordered not to open it to inspect the cargo, or to a vehicle loaded in a manner that makes inspecting the cargo impracticable. That is consistently stated by compliance sources, but we have not confirmed the wording against the regulation text — read 392.9(b) at ecfr.gov before relying on it.

7.3 AT THE SCALE

- **THE INSPECTION PRODUCES A REPORT — AND IT COMES BACK TO YOU**
  Under 49 CFR 396.9, the carrier must review the roadside inspection report, correct anything noted, and certify the corrections back to the issuing agency. That is a deadline, not a courtesy.

- **OUT OF SERVICE MEANS STOPPED — NOT "DRIVE CAREFULLY TO THE SHOP"**
  A vehicle declared out of service may not be operated until the defect is corrected. Moving it anyway is a violation carrying weight far beyond the original problem.

- **THE OFFICER JUDGES THE SYSTEM, NOT THE OUTCOME — IN FRONT OF THEM**
  Count, ratings, condition, edge protection, and arrangement. That the load has already travelled four states intact is not evidence that it was secured correctly.

IT FOLLOWS YOU AFTERWARDS — INTO YOUR PUBLIC RECORD
Inspection outcomes feed the federal safety systems that brokers and insurers look at. A pattern of securement violations costs loads and premium long after the citation is paid.

THE OUT-OF-SERVICE CRITERIA ARE A SEPARATE DOCUMENT

Inspectors apply the North American Standard Out-of-Service Criteria, published by the Commercial Vehicle Safety Alliance and revised annually. It sets out which violations put a vehicle out of service rather than simply generating a citation. It is a different document from the regulations in this guide, it changes each year, and if you want to know exactly where the out-of-service line falls on a securement defect, that is the source to consult.

7.4 TURNING IT INTO A ROUTINE

Nothing in this section is a regulation. It is what makes the regulations survivable on a working day.

BEFORE THE SEASON

- **INVENTORY AND RATE YOUR KIT ONCE**
  Write down the working load limit of every chain, strap, binder and ratchet, and the lowest rating in each assembly.
- **KNOW YOUR TWO NUMBERS**
  What your normal kit gives you crossed, and what it gives you same-side. Those are the two figures every decision runs on.
- **RETIRE THE UNMARKED**
  Anything without a legible rating is counted at the lowest default. Replace it rather than argue about it.
- **BUY EDGE PROTECTION THAT CRUSHES WELL**
  It must resist abrasion, cutting and crushing. Most cheap options fail the third.

EVERY LOAD

- **ASK WHICH SECTION GOVERNS**
  General rules, or one of the eleven commodity sections. Get this wrong and everything after it is wrong too.
- **DO BOTH CALCULATIONS**
  Minimum tiedown count, and aggregate working load limit. Whichever demands more, wins.
- **BLOCK THE FRONT WHERE YOU CAN**
  It halves the required count on long articles and it is free.
- **PHOTOGRAPH THE FINISHED LOAD**
  Not required by anything. Invaluable when a receiver claims the load shifted in transit.

THE THREE QUESTIONS THAT CATCH MOST ERRORS

Can it roll? Chocks, wedges, cradles, and the requirement that they be held by more than friction.
Can it tip? The ratio rules, the diagonal tiedowns, the front end structure.
Can it slide forward? The 0.8 g direction — blocking, friction mats, and the reason the front of a load gets the most attention.

Nearly every rule in Subpart I is answering one of those three. When a situation is not obviously covered, work out which one is unaddressed.

7.5 WHO TO CONTACT

FEDERAL MOTOR CARRIER SAFETY ADMINISTRATION

Phone	1-800-832-5660
Hours	Monday to Friday, 8 AM to 8 PM Eastern
Ask	Which commodity section applies to an unusual load you are being offered

READ THE RULES YOURSELF

The whole subpart	49 CFR Part 393, Subpart I, at ecfr.gov
The driver's duty	49 CFR 392.9
Out-of-service line	Commercial Vehicle Safety Alliance — North American Standard Out-of-Service Criteria

49 CFR 392.9 | 49 CFR 393.100–393.136 | 49 CFR 396.9

7.6 THE WHOLE THING, IN ONE LIST

Every tiedown, binder and anchor point carries a legible working load limit

You know the lowest-rated component in each assembly you use

You count same-side tiedowns at half and cross-side tiedowns at full

You satisfy both the minimum count and the 50 percent aggregate rule, separately

Anything that could roll is chocked, and the chocks are held by more than friction

Edge protection is in place wherever a tiedown crosses cargo, and it resists crushing

No tiedown contains a knot, and every binder still adjusts

You have checked whether a commodity-specific section governs your load

Tarps, tailboards, doors and the spare tire are secured

Nothing obstructs your view, your movement, or the exit

You inspect within the first 50 miles, and again every 3 hours or 150 miles

You know where to read the rule yourself when someone tells you otherwise

WHAT YOU ACTUALLY HAVE NOW

Cargo securement is the violation small carriers collect most, and almost never because anyone was careless. It is arithmetic — ratings counted wrong, straps counted at double their value, a commodity rule nobody knew existed. All of it is public, free, and written down, and none of it is hidden. It is simply scattered across a subpart of federal regulation in language written for inspectors rather than for the person buying their first flatbed.

Every figure in this guide can be checked at **ecfr.gov**. If anyone tells you something that contradicts what you read there, the regulation wins. That is the whole advantage, and now it is yours.